

# Jinmin Wang (王晋民)

Email: jinmin@amss.ac.cn

Homepage: jinminwang.github.io

## Research Interests.

$K$ -theory, index theory and noncommutative geometry

## Education.

|           |                                           |
|-----------|-------------------------------------------|
| 2015–2020 | Ph.D. of Mathematics, Fudan University    |
| 2011–2015 | Bachelor of Mathematics, Fudan University |

## Positions.

|              |                                                                            |
|--------------|----------------------------------------------------------------------------|
| 2024–current | Assistant Professor, Institute of Mathematics, Chinese Academy of Sciences |
| 2021–2024    | Visiting Assistant Professor, Texas A&M University                         |
| 2020–2021    | Research Assistant, Fudan University                                       |

## Publications.

- (with S. Cecchini, Z. Xie and B. Zhu) Scalar curvature rigidity of the four-dimensional sphere, to appear in *Math Annalen*.
- (with Z. Xie, B. Zhu, and X. Zhu) Positive scalar curvature meets Ricci limit spaces. *manuscripta mathematica*, 175(3), 943-969.
- (with Z. Xie) Scalar curvature rigidity of degenerate warped product spaces. *Transactions of the American Mathematical Society, Series B*, 12(01), 1-37.
- (with X. Chen, Z. Xie, and G. Yu) Delocalized eta invariants, cyclic cohomology and higher rho invariants, *Annals of K-Theory* 8-4 (2023), 531–608. DOI 10.2140/akt.2023.8.531
- (with Z. Xie and G. Yu)  $l^1$ -higher index,  $l^1$ -higher rho invariant and cyclic cohomology, *Cyclic Cohomology at 40: achievements and future prospects* (2021), 535-579.
- (with Z. Xie and G. Yu) Decay of scalar curvature on uniformly contractible manifolds with finite asymptotic dimension, *Communications on Pure and Applied Mathematics*, 77 (2024), no. 1, 372–440.

- (with Z. Xie and G. Yu) A proof of Gromov's cube inequality on scalar curvature. *Journal of Differential Geometry*, 128(3), 1285-1300.
- (with Z. Xie and G. Yu) Approximations of delocalized eta invariants by their finite analogues, *L'Enseignement mathématique*.
- (with H. Liu) On localized signature and higher rho invariant of fibered manifolds. *J. Noncommut. Geom.* 15 (2021), no. 3, pp. 919–949.

### Preprints.

- (with M. Jovanovic) A scalar-mean curvature comparison theorem for manifolds with iterated conical singularities, arXiv:2506.24059
- (with Z. Xie, G. Yu, B. Zhu)  $\ell^p$ -coarse Baum-Connes conjecture for  $\ell^q$ -coarse embeddable spaces, arXiv:2411.15070
- (with Q. Ma, G. Yu and B. Zhu) Bounding the A-hat genus using scalar curvature lower bounds and isoperimetric constants, arXiv:2410.08866
- (with B. Zhu and Z. Wang) Scalar-mean rigidity theorem for compact manifolds with boundary, arXiv:2409.14503
- (with B. Zhu) Sharp bottom spectrum and scalar curvature rigidity, arXiv:2408.08245
- (with Z. Xie) Scalar curvature rigidity of spheres with subsets removed and  $L^\infty$  metrics, arXiv:2407.21312
- (with Z. Xie, G. Yu and B. Zhu) Filling Radius, Quantitative  $K$ -theory and Positive Scalar Curvature, arXiv:2311.15347.
- (with Z. Xie) Dihedral rigidity for submanifolds of warped product manifolds, arXiv:2303.13492.
- (with Z. Xie) Rigidity of strictly convex domains in Euclidean spaces, arXiv:2207.05731.
- (with Z. Xie) On Gromov's flat corner domination conjecture and Stoker's conjecture, arXiv:2203.09511.
- (with Z. Xie and G. Yu) On Gromov's dihedral extremality and rigidity conjectures, arXiv:2112.01510.

## Invited Conference and Seminar Talks.

### 2025.

- Geometry and Convergence Program, Simons Center (September)
- Special Week on Operator Algebras 2025, Shanghai Institute for Mathematics and Interdisciplinary Sciences (August)
- 4th Australia-China-Japan-Singapore-US Index Theory Conference – Analysis and Geometry on Manifolds, National University of Singapore (August)
- Work shop on noncommutative geometry and operator algebra, Sichuan University (May)

### 2024.

### 2023.

- Analysis seminar, University at Buffalo (November)
- Noncommutative Geometry Festival 2023, Washington University in St. Louis (June)
- Brazos Analysis Seminar, Texas A&M University (April)
- Global Noncommutative Geometry Seminar, America (March, online)
- Joint mathematics meeting, AMS Special Session on Riemannian Manifolds with Lower Scalar Curvature Bounds, Boston (January)

### 2022.

- Wabash mini workshop, University of Indiana (November)
- K-theory of operator algebras and its applications to geometry and topology, University of Puerto Rico (August)
- Young Mathematicians in Noncommutative Geometry and Analysis, Texas A&M University (August)
- Recent Advances on Scalar Curvature Problems, Simons Center, Stony Brook (June)
- Great Plains Operator Theory Symposium, Washington University in St. Louis (May)
- Spring 2022 Workshop on NCG and Geometric Analysis (April, online)
- Not Only Scalar Curvature Seminar (April, online)

### 2021.

- Global Noncommutative Geometry Seminar, Europe (December, online)

2020.

- Workshop on Noncommutative Geometry and Quantization, Sichuan University (November)

2019.

- National Conference on Operator Theory and Operator Algebra, Zhejiang University, Hangzhou (November)
- Special Week on Operator Algebras 2019, East China Normal University, Shanghai (June)
- Noncommutative Geometry Festival 2019, Washington University in St. Louis (April)

### **Teaching Experience.**

|             |                                                               |
|-------------|---------------------------------------------------------------|
| 2024 fall   | Functional Analysis II                                        |
| 2023 fall   | Math 251, Calculus III                                        |
| 2023 spring | Math 323, Linear Algebra (for mathematics and science majors) |
| 2022 fall   | Math 251, Calculus III                                        |
| 2022 spring | Math 304, Linear Algebra (for engineering majors)             |
| 2021 fall   | Math 251, Calculus III                                        |